

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Online Token Generation System

Gujarat State Government

- **Project Title:** Online Token Generation System
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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe in detail the functional and non-functional requirements of the Online Token Generation System for Government Offices in Gujarat State.

This document serves as a reference for developers, project supervisors, system architects, and stakeholders. It defines system behavior, performance expectations, security requirements, and design constraints to ensure proper development and implementation.

The SRS will act as a formal agreement between stakeholders and developers regarding system functionality.

1.2 Scope

The Online Token Generation System is a web-based application designed to reduce long waiting queues in government offices by providing online token booking and document tracking services.

The system will:

- Allow citizens to book tokens online
- Provide real-time token tracking
- Generate unique inward numbers after document submission
- Enable staff to manage queue efficiently
- Allow administrators to manage offices and services
- Improve transparency and accountability

The system will not:

- Replace official government verification processes
- Store biometric or Aadhaar data (in current version)
- Process online payments (future enhancement)

The application is designed for implementation at the Gujarat State level and can be expanded further.

1.3 Definitions, Acronyms, Abbreviations

SRS – Software Requirements Specification

RTO – Regional Transport Office

OTP – One Time Password

JWT – JSON Web Token

RBAC – Role-Based Access Control

UI – User Interface

API – Application Programming Interface

1.4 References

The following references were used in preparing this document:

- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications
- Django Official Documentation (<https://docs.djangoproject.com>)
- ReactJS Official Documentation (<https://react.dev>)
- Database System Concepts – Silberschatz, Korth, Sudarshan

1.5 Document Overview

This document is structured into multiple sections describing the overall system architecture, system features, functional requirements, non-functional requirements, system models, and future enhancements.

Section 1 introduces the system.

Section 2 describes the overall system perspective.

Section 3 explains system modules.

Section 4 describes interface requirements.

Section 5 defines functional requirements in detail.

Section 6 defines non-functional requirements.

Section 7 includes system models such as DFD, Use Case, and ER diagrams.

Section 8 describes future enhancements.

Section 9 concludes the document.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Online Token Generation System is a web-based client-server application.

The system architecture consists of:

- Frontend Layer: Developed using React.js for user interface
- Backend Layer: Developed using Django REST Framework for business logic
- Database Layer: MySQL or PostgreSQL for data storage
- Blockchain Layer (Conceptual): Used for timestamp verification

The system follows a modular architecture and can be integrated with future government platforms.

2.2 Product Functions

The major functions of the system include:

- User registration and authentication
- Role-based access control
- Online token booking
- Unique token number generation
- Real-time queue status updates
- Inward number generation for document tracking
- Application status tracking
- Administrative management of offices and services
- Reporting and monitoring

2.3 User Classes and Characteristics

Citizen

- Basic knowledge of internet usage
- Books tokens online
- Tracks token and application status

Employee

- Government office staff
- Manages token queue
- Updates application status

Admin

- Manages office services
- Monitors token generation
- Manages staff accounts

Super Admin

- Has full system control
- Manages districts, offices, and system configurations

2.4 Operating Environment

The system operates in the following environment:

- Operating Systems: Windows, Linux, macOS
- Web Browsers: Google Chrome, Mozilla Firefox, Microsoft Edge
- Backend Server: Python (Django Framework)
- Database Server: MySQL / PostgreSQL
- Hosting Environment: Cloud-based or Local Government Server

2.5 Design Constraints

- The system must ensure data privacy and security.
- Token numbers must be unique and non-duplicated.
- System must handle multiple concurrent users.
- Role-based access must be strictly enforced.
- Government compliance guidelines must be followed.

2.6 Assumptions and Dependencies

- Users have internet connectivity.
- Government offices have computers and internet access.
- Backend server infrastructure will be available.
- System depends on database availability.
- OTP service provider will be integrated in future implementation.

3. SYSTEM FEATURES

This section describes the major modules of the Online Token Generation System in detail, along with their functional behavior.

3.1 User Authentication Module

Description

This module handles user registration, login, and role-based access control. It ensures that only authorized users can access system functionalities.

Functional Requirements:

FR-1: The system shall allow new users to register using Name, Email, Mobile Number, and Password.

FR-2: The system shall validate user input before registration.

FR-3: The system shall verify user identity using OTP (future integration).

FR-4: The system shall encrypt passwords before storing them in the database.

FR-5: The system shall allow registered users to log in using email/mobile and password.

FR-6: The system shall implement role-based access control (Citizen, Employee, Admin, Super Admin).

FR-7: The system shall redirect users to their respective dashboards after login.

FR-8: The system shall allow users to log out securely.

3.2 Token Booking Module

Description

This module enables citizens to book tokens online for various government services.

Functional Requirements

FR-9: The system shall allow users to select District, Taluka, and Office.

FR-10: The system shall display available services for selected office.

FR-11: The system shall allow users to select date and time slot.

FR-12: The system shall generate a unique token number.

FR-13: The token number format shall include state code and serial number (e.g., GJ01-0023).

FR-14: The system shall prevent duplicate token generation.

FR-15: The system shall store token details in the database.

FR-16: The system shall generate a downloadable token receipt.

FR-17: The system shall show estimated waiting time.

3.3 Queue Management Module

Description

This module is used by employees to manage and update token status in real-time.

Functional Requirements

FR-18: The system shall allow employees to view all tokens for the day.

FR-19: The system shall display current token being served.

FR-20: The system shall allow employee to call next token.

FR-21: The system shall allow employee to mark token as Completed, Cancelled, or Absent.

FR-22: The system shall update token status in real-time.

FR-23: The system shall maintain token history for reporting purposes.

3.4 Inward Number & Document Tracking Module

Description

This module generates an inward number after document submission and allows application tracking.

Functional Requirements

FR-24: The system shall generate a unique inward number after document submission.

FR-25: The inward number shall follow format: IN/District/Office/Year/Serial.

FR-26: The system shall store inward number in the database.

FR-27: The system shall update application status (Received, Under Verification, Approved, Rejected, Issued).

FR-28: The system shall allow citizens to track application using inward number.

FR-29: The system shall maintain a status change log with timestamps.

FR-30: The system shall prevent modification of inward number once generated.

3.5 Admin Management Module

Description

This module allows administrative control over offices, services, and system configurations.

Functional Requirements

FR-31: The system shall allow Super Admin to add districts.

FR-32: The system shall allow Super Admin to register new offices.

FR-33: The system shall allow Admin to add or remove services.

FR-34: The system shall allow Admin to create employee accounts.

FR-35: The system shall allow Admin to generate daily reports.

FR-36: The system shall allow Super Admin to monitor system activity.

4. External Interface Requirements

This section describes how the Online Token Generation System interacts with users, hardware, software, and communication interfaces.

4.1 User Interface

The system shall provide a responsive and user-friendly web interface.

User Interface Requirements:

- The system shall provide a clean and intuitive navigation menu.
- The interface shall be responsive and accessible on desktop, tablet, and mobile devices.
- The system shall provide separate dashboards for:
 - Citizen
 - Employee
 - Admin
 - Super Admin
- The system shall display token status and application tracking clearly.
- The system shall use proper form validation messages.
- The system shall provide confirmation messages after token booking.
- The interface shall follow government-style color themes (blue and white).

The UI will be developed using React.js and modern CSS design standards.

4.2 Hardware Interface

The system does not require specialized hardware.

Hardware Requirements:

- Server machine for hosting backend application
- Desktop computers or laptops for office staff
- Citizen devices (mobile phones, tablets, desktops)
- Internet connectivity

Minimum Server Requirements (Recommended):

- Processor: Intel i5 or equivalent
- RAM: 8GB minimum
- Storage: 100GB
- Reliable internet connection

4.3 Software Interface

The system shall interact with the following software components:

Backend Framework:

- Django REST Framework (Python)

Database:

- MySQL or PostgreSQL

Frontend:

- React.js

External Services (Future Scope):

- OTP verification service
- SMS notification service
- Email notification service

The system shall use REST APIs to communicate between frontend and backend.

4.4 Communication Interface

The system shall communicate over secure internet protocols.

Communication Requirements:

- All communication shall use HTTP/HTTPS protocol.
- Sensitive data transmission shall use SSL encryption.
- The system shall support RESTful API communication.
- The system shall handle concurrent client requests efficiently.

5. Functional Requirements (Detailed)

The functional requirements describe the specific behavior and operations of the Online Token Generation System.

5.1 User Registration & Authentication Requirements

FR-1: The system shall allow new users to register using Name, Email, Mobile Number, and Password.

FR-2: The system shall validate user input fields before submission.

FR-3: The system shall prevent duplicate registration using the same email or mobile number.

FR-4: The system shall encrypt user passwords before storing them in the database.

FR-5: The system shall allow registered users to log in using valid credentials.

FR-6: The system shall implement role-based access control (Citizen, Employee, Admin, Super Admin).

FR-7: The system shall redirect users to their respective dashboards after successful login.

FR-8: The system shall allow users to securely log out.

5.2 Token Booking Requirements

FR-9: The system shall allow citizens to search for offices by District and Taluka.

FR-10: The system shall display available services for the selected office.

FR-11: The system shall allow citizens to select date and time slot.

FR-12: The system shall generate a unique token number for each booking.

FR-13: The system shall ensure token number uniqueness.

FR-14: The system shall store token details in the database.

FR-15: The system shall display estimated waiting time.

FR-16: The system shall generate a downloadable token receipt.

5.3 Queue Management Requirements

FR-17: The system shall allow employees to view all tokens generated for the day.

FR-18: The system shall display the current token being served.

FR-19: The system shall allow employees to call the next token.

FR-20: The system shall allow employees to update token status (Completed, Cancelled, Absent).

FR-21: The system shall update token status in real-time.

FR-22: The system shall maintain token service history.

5.4 Inward Number & Document Tracking Requirements

- FR-23: The system shall generate a unique inward number after document submission.
- FR-24: The inward number shall follow a predefined format.
- FR-25: The system shall store inward number details in the database.
- FR-26: The system shall allow application status updates (Received, Under Verification, Approved, Rejected, Issued).
- FR-27: The system shall allow citizens to track application status using inward number.
- FR-28: The system shall maintain status logs with timestamps.
- FR-29: The system shall prevent modification of inward number once generated.

5.5 Administrative Requirements

- FR-30: The system shall allow Super Admin to add districts and offices.
- FR-31: The system shall allow Admin to manage services.
- FR-32: The system shall allow Admin to create employee accounts.
- FR-33: The system shall allow Admin to generate reports.
- FR-34: The system shall allow Super Admin to monitor system activity.

6. Non-Functional Requirements

Non-functional requirements describe system performance, security, usability, reliability, and scalability characteristics.

6.1 Performance Requirements

- NFR-1: The system shall generate a token within 2 seconds under normal load conditions.
- NFR-2: The system shall respond to user actions within 3 seconds.
- NFR-3: The system shall support at least 1000 concurrent users.
- NFR-4: The system shall handle a minimum of 500 token bookings per hour without performance degradation.
- NFR-5: The system shall update queue status in real-time or within 1 second delay.

6.2 Security Requirements

- NFR-6: The system shall encrypt user passwords using secure hashing algorithms (e.g., bcrypt).
- NFR-7: The system shall implement role-based access control (RBAC).
- NFR-8: The system shall use HTTPS protocol for secure communication.

NFR-9: The system shall prevent SQL injection and cross-site scripting (XSS) attacks.

NFR-10: The system shall restrict unauthorized access to administrative features.

NFR-11: The system shall maintain activity logs for auditing purposes.

6.3 Reliability Requirements

NFR-12: The system shall maintain 99% uptime availability.

NFR-13: The system shall automatically save token and application data to prevent data loss.

NFR-14: The system shall recover gracefully from system crashes.

NFR-15: The system shall maintain data consistency in case of concurrent access.

6.4 Usability Requirements

NFR-16: The system shall provide a simple and user-friendly interface.

NFR-17: The system shall be responsive on desktop, tablet, and mobile devices.

NFR-18: The system shall display clear error and confirmation messages.

NFR-19: The system shall allow first-time users to complete token booking without technical assistance.

NFR-20: The interface shall follow consistent design standards.

6.5 Scalability Requirements

NFR-21: The system shall support expansion to all districts of Gujarat.

NFR-22: The system architecture shall allow addition of new offices without major redesign.

NFR-23: The database shall support increasing records without affecting performance.

NFR-24: The system shall support future integration with mobile applications.

7. System Models

This section describes the system using different modeling techniques such as Use Case Diagram, Data Flow Diagram (DFD), and Entity Relationship (ER) Diagram.

These models help in understanding system functionality, data flow, and database structure.

7.1 Use Case Diagram

Description: The Use Case Diagram represents the interaction between different users (actors) and the system. It shows how users interact with the system functionalities.

Actors

- Citizen
- Employee
- Admin
- Super Admin

Use Cases

For Citizen:

- Register
- Login
- Book Token
- View Token Status
- Track Application

For Employee:

- View Tokens
- Call Next Token
- Update Token Status
- Generate Inward Number

For Admin:

- Manage Services
- Manage Employees
- View Reports

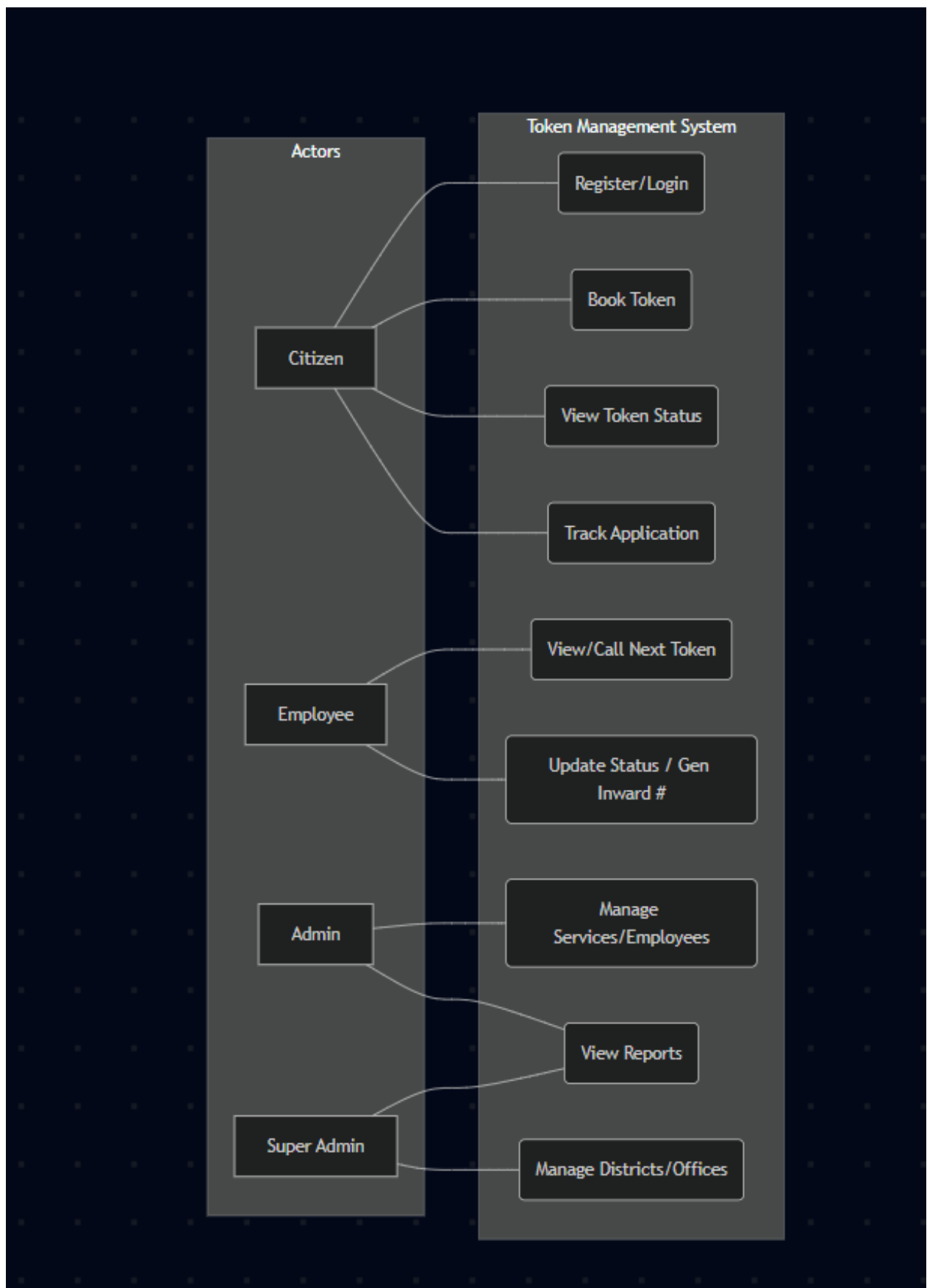
For Super Admin:

- Manage Districts
- Add Offices
- Monitor System

Purpose

The Use Case Diagram helps to visualize system functionality from the user's perspective and defines user interactions clearly.

Use Case Diagram:



7.2 Data Flow Diagram (DFD)

Description

The Data Flow Diagram (DFD) represents how data moves through the system and how different processes interact with data sources.

Level 0 DFD (Context Diagram)

The entire system is represented as a single process interacting with external entities:

- Citizen provides input (token booking, tracking request)
- System processes data
- Database stores token and application data
- Employee/Admin updates system data

Level 1 DFD

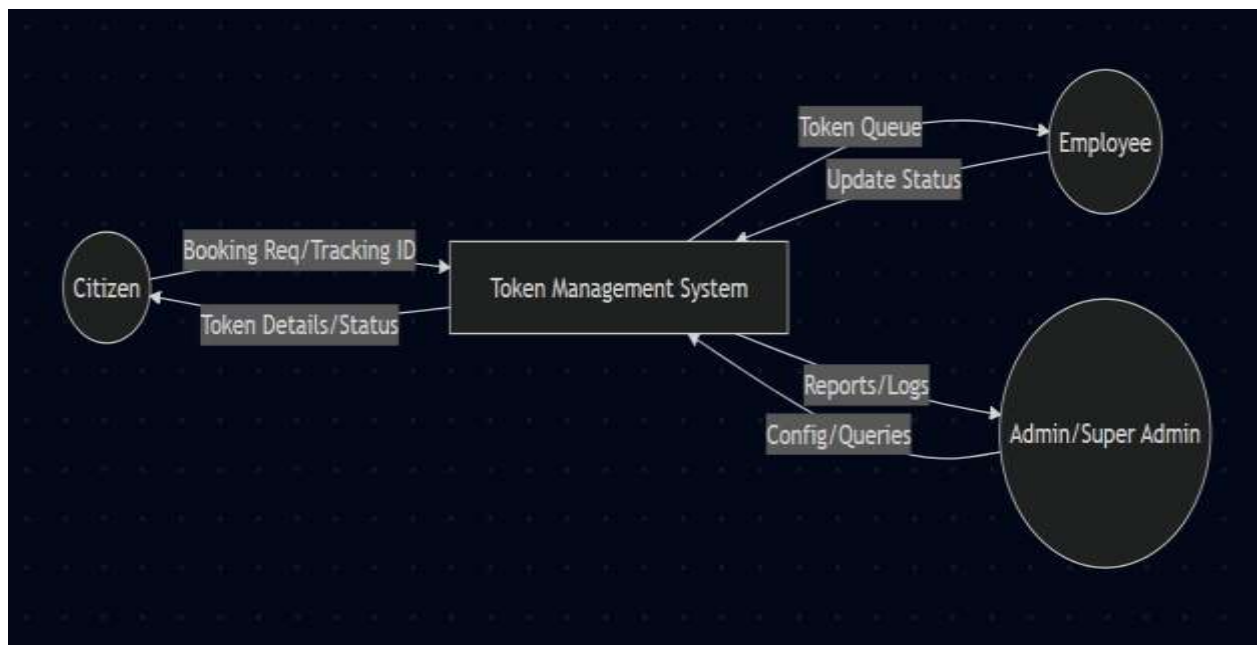
The system is divided into major processes:

- User Authentication Process
- Token Booking Process
- Queue Management Process
- Inward Number Generation Process
- Application Tracking Process

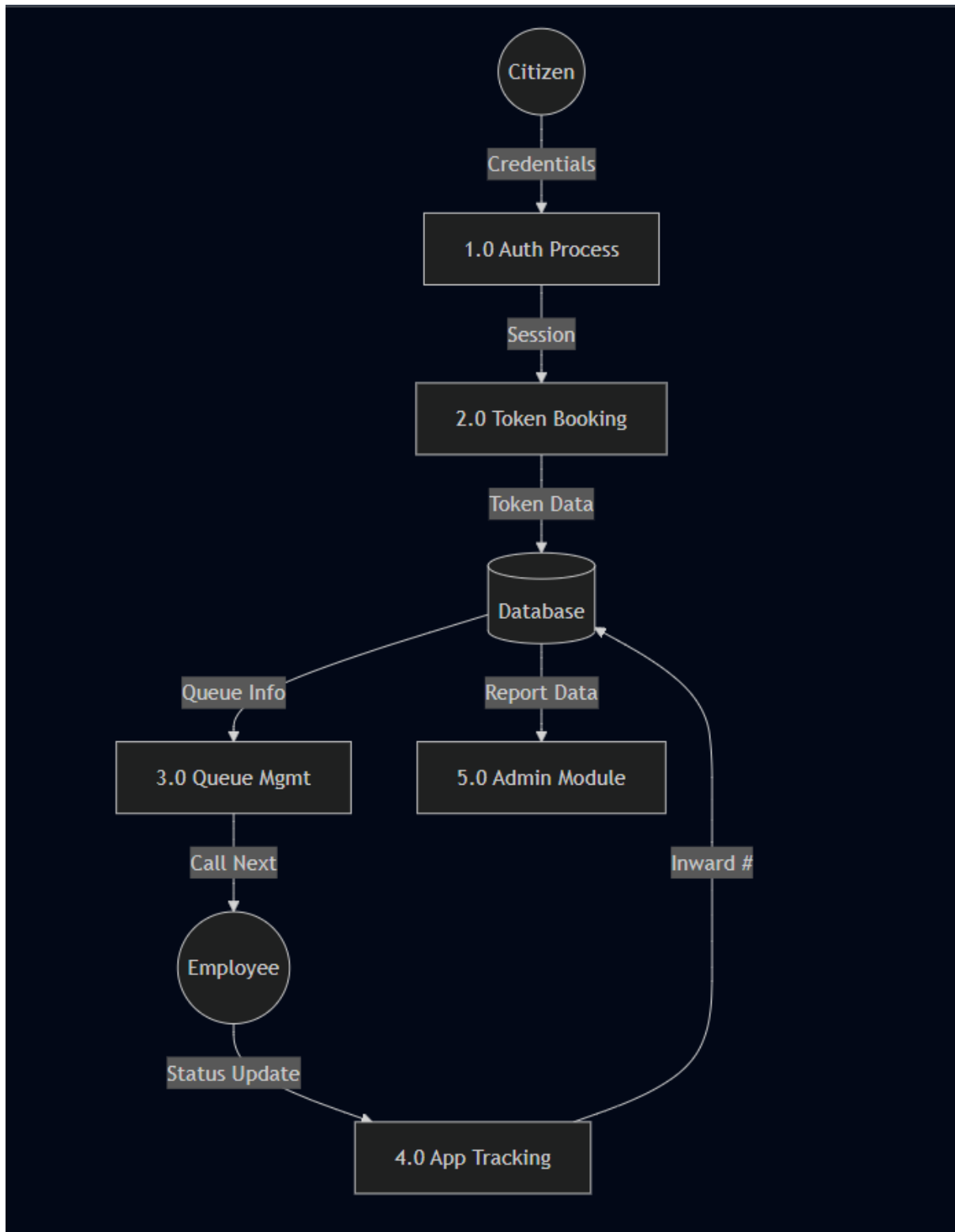
Purpose

The DFD helps in understanding system workflow, data movement, and processing logic.

DFD Level 0 diagram:



DFD Level 1 diagram:



7.3 Entity Relationship (ER) Diagram

Description:

The ER Diagram represents the database structure of the system, including entities, attributes, and relationships.

Main Entities

- User
- Office
- Service
- Token
- Application
- Inward Number
- Status Log

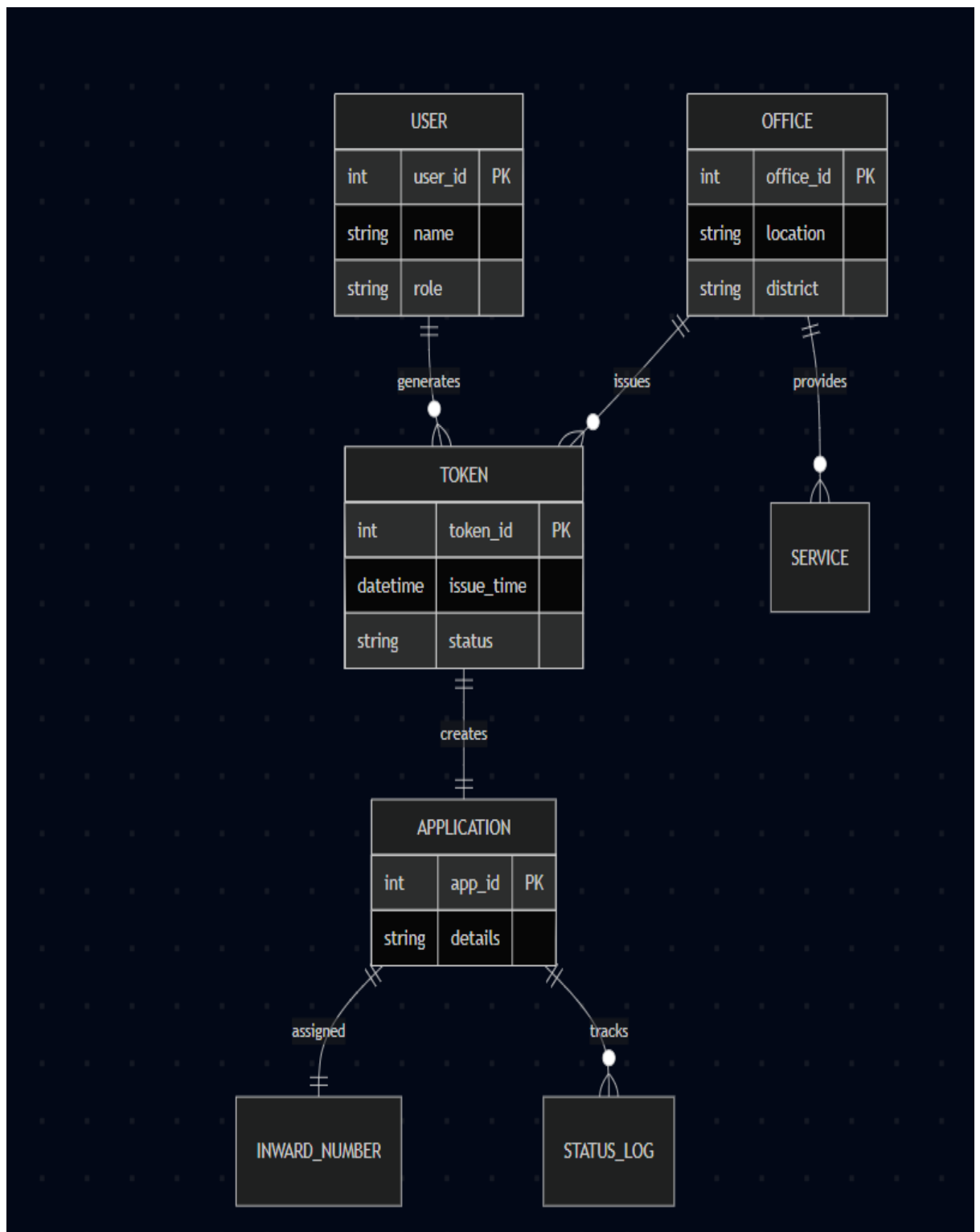
Relationships

- A User can generate multiple Tokens
- A Token belongs to one Office
- A Token is associated with one Application
- An Application has one Inward Number
- An Application can have multiple Status Logs
- An Office provides multiple Services

Purpose

The ER Diagram helps in designing the database and understanding relationships between different entities.

ER Diagram:



8 Future Enhancements

The current system provides core functionality for token generation and document tracking. However, several enhancements can be implemented in future versions to improve system performance, usability, and scalability.

- Integration with **OTP-based authentication** for enhanced security
 - Development of a **mobile application (Android/iOS)** for easier access
 - Integration with **SMS and Email notification systems** for token updates
 - Aadhaar-based verification for user authentication (future scope)
 - Implementation of **AI-based waiting time prediction**
 - Real-time display system for tokens inside government offices
 - Multi-language support for better accessibility across Gujarat
 - Advanced blockchain integration for secure and immutable record keeping
 - Online payment integration for paid government services
- These enhancements will further improve the efficiency, accessibility, and reliability of the system.

9 Conclusion:

The Online Token Generation and Document Tracking System aims to modernize the traditional manual process used in government offices by introducing a digital and automated solution. The system reduces waiting time, improves transparency, and enhances the overall efficiency of service delivery.

By enabling online token booking, real-time queue management, and inward number-based document tracking, the system provides a structured and user-friendly approach for both citizens and government staff. The use of modern web technologies ensures scalability, security, and ease of use.

The proposed system can significantly reduce crowding in offices, minimize manual errors, and improve accountability. It also lays the foundation for future enhancements such as mobile applications, AI integration, and secure data verification mechanisms. Overall, the system contributes to better governance and improved citizen experience.